

SANMAY DAS

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EDUCATION

Master of Science in Computer Science

University of California, Riverside

GPA: 3.76/4.0

Relevant Courses: Artificial Intelligence, Big Data Management, Cloud Computing, GPU Programming, Advanced Computer Vision, Machine Learning, Compiler Construction, Information Retrieval & Web Search, Design & Analysis of Algorithms

September 2024 – March 2026

Riverside, CA

Bachelor of Technology in Information Technology

KIIT University

August 2017 – May 2021

Bhubaneswar, India

PUBLICATIONS

Zhuocheng Shang, **Sanmay Das**, and Ahmed Eldawy. "Benchmarking Geospatial Foundation Models for Agriculture Applications." Accepted to the 34th ACM SIGSPATIAL International Conference on Advances in Geographic Information Systems (**SIGSPATIAL '26**), Riverside, California, 2026 | <https://arxiv.org/abs/2606.29664>

RESEARCH EXPERIENCE

Graduate Researcher

University of California, Riverside

September 2025 – Present

Riverside, CA

- Designed geographically disjoint training, validation, and test splits across four U.S. states to evaluate spatial generalization without regional overlap
- Built reproducible PyTorch pipelines to fine-tune and evaluate Prithvi-EO, SpectralGPT, and SatMAE for multitemporal crop semantic segmentation and change detection using A100 GPUs and SLURM
- Processed 1 TB of multitemporal Sentinel-2 imagery into 500K+ training chips through a reproducible ETL pipeline
- Found that geographic distribution shift disproportionately affected minority crops, with crop-class mIoU declining by up to 71%, while aggregate mIoU concealed severe class-level failures

TEACHING EXPERIENCE

Grader – CS 205: Artificial Intelligence

University of California, Riverside

January 2025 – March 2025

Riverside, CA

- Evaluated assignments and examinations for approximately 100 students on knowledge representation, automated reasoning, predicate logic, planning, and probabilistic reasoning
- Applied consistent grading rubrics and provided clear feedback while coordinating with the instructor to resolve ambiguous solutions and maintain fair assessment standards

PROFESSIONAL EXPERIENCE

Student Assistant

University of California, Riverside

June 2025 – August 2025

Riverside, CA

- Served geospatial queries for [FutureFarmNow](#) ([Live Site](#)) by deploying Python REST APIs with Flask and Unicorn
- Achieved an 80% cache hit rate by engineering distributed data workflows with caching and SQLite
- Reduced deployment time by 75% by architecting automated CI/CD with Git, GitHub Actions, and PyTest gates

Software Engineer

LTIMindtree

June 2021 – November 2023

Bhubaneswar, India

- Lowered API failures by 50% by delivering C#/.NET REST APIs backed by SQL Server relational databases
- Cut deployment risk by 25% via CI/CD with unit, integration, and regression testing for a .NET MAUI app
- Reduced debugging effort by 40% by automating crash-log processing and exception triage with Python

PROJECTS

[Eaton Fire Satellite Analysis](#) | [Scala](#), [GeoTrellis](#), [Landsat 8/9](#), [GDAL](#), [ImageMagick](#), [FFmpeg](#)

- Developed a multitemporal Earth Observation workflow visualizing vegetation change from the 2025 Eaton Fire
- Implemented NDVI computation in Scala with GeoTrellis, applying reflectance scaling and no-data masking
- Automated conversion of NDVI rasters into time-series GIF and MP4 outputs via GDAL, ImageMagick, and FFmpeg

[CityCrimes.io](#) - [GitHub](#) | [Go](#), [GraphQL](#), [PostgreSQL](#), [React](#), [TypeScript](#), [Docker](#), [Groq](#), [Nomic AI](#)

- Processed 10.8M+ crime records via a Go/GraphQL/PostgreSQL platform with an LLM-powered RAG pipeline

- Enabled natural-language querying by integrating Groq LLaMA-3.3-70B with Nomic vector search
- Delivered ~70 ms p50 GraphQL query latency by tuning Go API resolvers over PostgreSQL

LLM Serving Benchmark | *Python, Kubernetes, Docker, vLLM, Ollama, Ray Serve, Prometheus, Grafana*

- Achieved 57% higher throughput and 38% lower tail latency by benchmarking vLLM, Ollama, and Ray Serve
- Deployed three **LLM** serving runtimes to a 3-node **Kubernetes** cluster with **Docker** containers
- Monitored inference latency and throughput in production with **Prometheus** and **Grafana** dashboards

CUDA Parallel Algorithms – Graduate Coursework | [Matrix Multiplication](#) | [Parallel Reduction](#) | CUDA C++

- Implemented a 16×16 shared-memory tiled matrix-multiplication kernel supporting square and rectangular matrices, including dimensions not aligned with the tile size
- Developed naive interleaved-addressing and optimized sequential-addressing parallel-reduction kernels using shared memory and thread synchronization
- Instrumented memory allocation, host–device data transfers, and kernel-execution timings and validated GPU results against sequential CPU reference implementations

TECHNICAL SKILLS

Programming: Python, Go, C#, Java, Scala, SQL, Bash, C, C++

Machine Learning: PyTorch, Hugging Face, scikit-learn, pandas, NumPy, TerraTorch

LLM Engineering: vLLM, Ollama, Ray Serve, LLM APIs, retrieval-augmented generation

Experiment Infrastructure: Linux, A100 GPU computing, SLURM, HPC, Kubernetes, Docker, Prometheus, Grafana

Software and Data Systems: Git, GitHub Actions, pytest, Flask, PostgreSQL, SQL Server, SQLite, Redis, GraphQL

Geospatial Processing: Rasterio, GDAL, GeoPandas, GeoTrellis

AWARDS

Silver Larry L. Sautter Award for Innovation (FutureFarmNow Team) | UC Tech Awards – 2026